ScholarX

Autonomous Scholarship Intelligence for Pakistani Students

A multi-agent, fully autonomous system designed to discover, fact-check, and plan fully funded global scholarships abroad according to your profile and preferences.

Contents

01. Brain (LLM Core)

High-order reasoning, task decomposition, and strategic planning powered by Gemini.

02. Tool Execution

Directly invokes external tools: web search (Tavily), crawlers (Firecrawl), and PDF parsers.

03. Autonomous Agency

Operates independently toward an objective without requiring step-by-step human prompts.

04. Memory Systems

Retains working context for live tasks, plus persistent database storage across searches.

05. Iterative Loop

Cycles continuously: Perceive → Think → Act → Observe → Verify until the goal is met.

06. Goal-Oriented

Optimizes every step to find eligible, verified, fully funded scholarship matches.

What is an AI Agent?

An autonomous software entity equipped with reasoning, memory, tools, and goals that operates in an iterative loop.

Agenda

Multi-Agent Systems

How specialized agents collaborate, challenge each other, and achieve the overarching scholarship agenda.

Specialized Division of Labor

Each agent masters one domain: Scouts discover, Eligibility Checkers audit, and Skeptics stress-test.

Shared State & Event Bus

Asynchronous pub/sub communication via Redis and LangGraph channels maintains system synchronization.

Adversarial Debate & Cross-Checking

The Skeptic agent actively attempts to invalidate claims, requiring primary official proof for approval.

Collective Agenda Convergence

All agent discoveries converge into a ranked, verified, fraud-free scholarship roadmap for the student.

SECTION BREAK: WORKFLOWS VS. AGENTS

Deterministic code meets probabilistic AI reasoning. Why ScholarX strictly requires both.

01. Deterministic Code

100% predictable with zero hallucination. Computes the 15 eligibility hard gates, currency conversion (PKR), and deadlines on the CPU with instant execution.

RULES, MATH & DEADLINES

02. Probabilistic Agents

Natural comprehension and adaptive search. Understands complex student CVs, discovers live opportunities, and challenges suspicious claims with LLM reasoning.

EXTRACTION & SKEPTIC AUDIT

03. The ScholarX Law

Deterministic workflows strictly guardrail AI agents. No LLM is ever allowed to guess a deadline or calculate PKR costs. Zero token hallucination on critical data.

ARCHITECTURAL GUARANTEE

The Problem

Every year, thousands of fully funded international seats reserved for Pakistan expire empty while families face opaque bureaucracy.

Why Scholarships Go Unclaimed

- **Hidden Quotas:** 40%+ of bilateral seats (Hungary, China CSC, Turkiye) expire unawares.
- **Cost Shock:** Surprise upfront PKR 150K+ fees for IELTS, visa, and legal stamps.
- **Predatory Agents:** Extorting PKR 150K–300K merely to file free public forms.

ScholarX is 100% free forever for every Pakistani student.

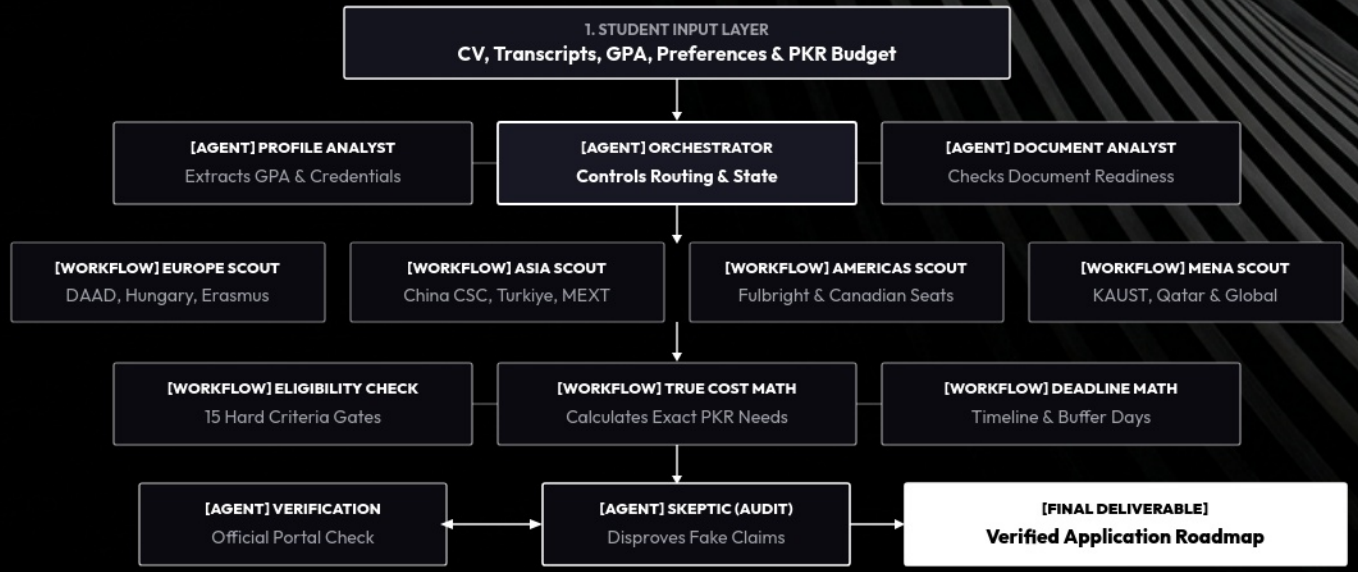


The Paperwork Maze

HEC HAT tests, IBCC equivalencies, and MOFA legal queues disqualify eligible candidates.

SYSTEM FLOWCHART: ALL AGENTS & WORKFLOWS

5 autonomous AI agents and 7 deterministic workflows orchestrated in one unified field.



THE STUDENT JOURNEY IN 4 STEPS

From uploading your CV to getting a verified application roadmap in under 2 minutes.

01

Profile Intake

2 MINUTES

Upload your CV or transcript. The AI parses GPA, degree type, publications, and test scores without tedious forms.

02

Global Scan

AUTONOMOUS

4 regional scouts crawl official government databases across Europe, Asia, Americas, and MENA in parallel.

03

Fact-Checking

STRICT AUDIT

Filters through 15 qualification gates, calculates true PKR expenses, and challenges claims with the Skeptic Agent.

04

Action Roadmap

READY TO APPLY

Get a ranked dossier with verified portal links, required HEC/IBCC steps, and dynamic deadline reminders.

~90s
Average Hunt Duration

15 Gates
Strict Rule Checks

PKR 0
Total Cost to Student

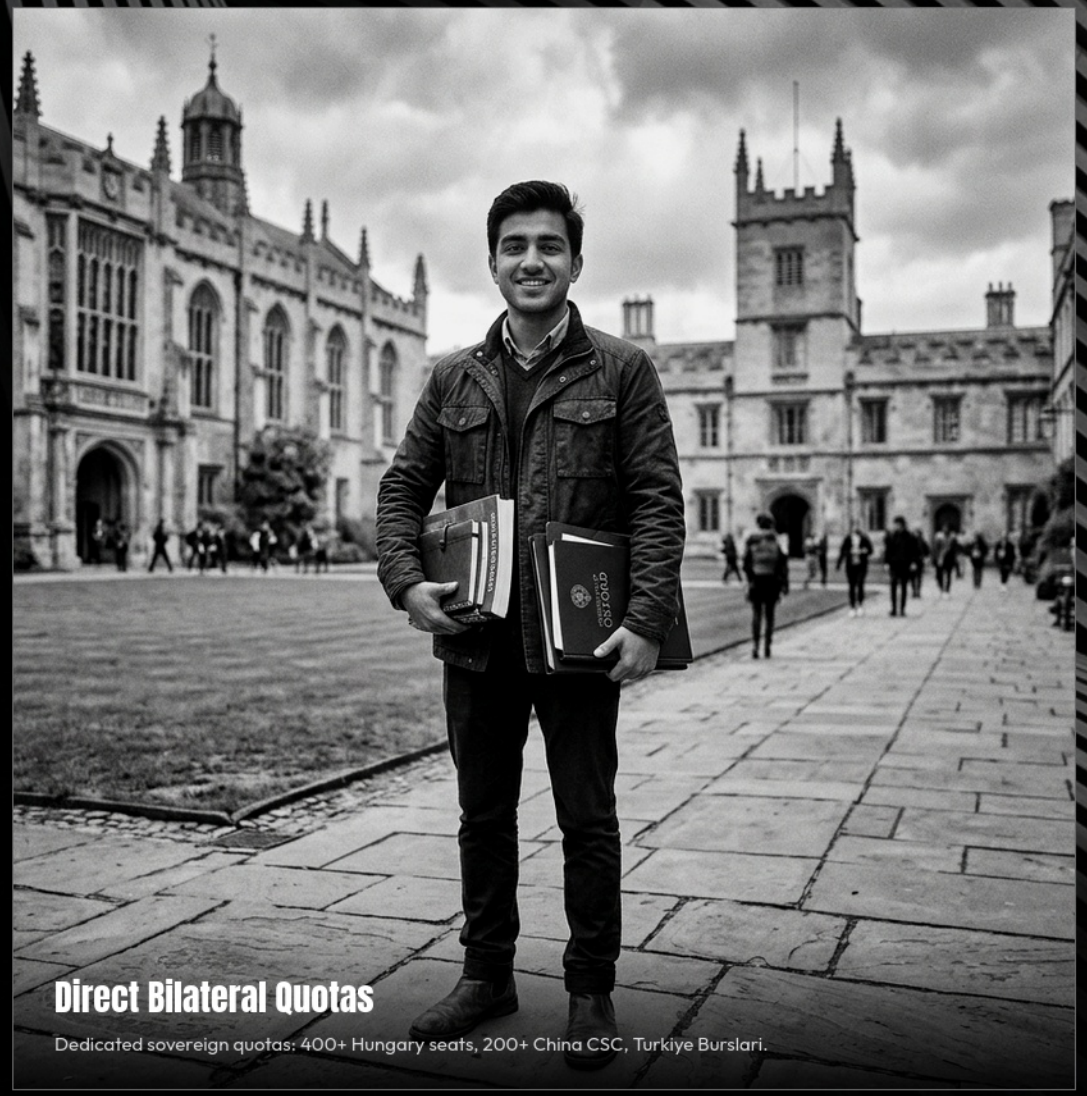
Direct official government links only. No sponsored or affiliate redirects.

Built for Pakistan

Engineered specifically around Pakistani academic credentials and real pre-departure cash realities.

MANDATORY PRE-DEPARTURE EXPENSE	TYPICAL COST (PKR)
IELTS / TOEFL Exam	**PKR 75,600 – 76,240**
HEC & IBCC Attestations	**PKR 12,000 – 18,000**
Embassy Visa & Biometrics	**PKR 28,000 – 45,000**
Medical Exam (Approved Clinic)	**PKR 22,000 – 35,000**
Total Readiness Budget:	**PKR 137K – 174K**

Financial Honesty: ScholarX factors upfront cash needs into match ranking so students are never blindsided.



Direct Bilateral Quotas

Dedicated sovereign quotas: 400+ Hungary seats, 200+ China CSC, Turkiye Burslari.

COMPLETE TECH STACK & ARCHITECTURE

Production-grade local architecture combining state-of-the-art Python orchestration and modern Next.js.

Frontend (UI)

- Next.js 15 (App Router)
- React 19 & TypeScript
- Tailwind CSS v4
- Framer Motion
- @xyflow/react (Graph)

PORT 3000

Backend Core

- Python 3.12+
- FastAPI (Async ASGI)
- LangGraph State Machine
- LangChain Core
- Pydantic v2 (Validation)

PORT 8000

Parsers & DB

- PyMuPDF & pdfplumber
- Tavily & Firecrawl APIs
- PostgreSQL + pgvector
- Redis 7 Pub/Sub
- SQLite Session Store

LOCAL DOCKER

LLMs & Failover

- Gemini 3.7 Flash (Primary)
- Gemini 3.0 Flash (Fallback)
- Groq gpt-oss-120b
- Multi-Key Key Rotation
- Zero Outage Guarantee

AUTO-ROTATING

RESILIENT MULTI-KEY FAILOVER: AUTOMATIC KEY ROTATION PREVENTS DISRUPTIONS

100%
FREE FOR ALL STUDENTS

400+
BILATERAL SEATS TRACKED

15/15
ELIGIBILITY GATES CHECKED

PKR 0
CONSULTANT COMMISSIONS

"A student's talent—not their wealth or connections—should determine their future."

ScholarX • Science Exhibition 2026 • Sovereign AI built with pride for Pakistan.

Autonomous Multi-Agent Intelligence

5 specialized AI agents + 7 deterministic Python workflows working in harmony.

Skeptical Proof Architecture

Every deadline and requirement audited against primary embassy portals.

Locally Deployable Stack

FastAPI backend, Next.js frontend, and persistent Redis/SQLite state.